higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N6

(8060056)

21 November 2019 (X-Paper)
09:00–12:00

Nonprogrammable calculators may be used.

This question paper consists of 5 pages, 2 addenda and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL SURVEYING N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches should be neat and clearly labelled.
5. Start each question on a NEW page.
6. Write neatly and legibly.

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QUESTION 1

Explain each of the following terms:

1.1 Back sight

1.2 Field book

1.3 Chaining

1.4 Chainage

1.5 Isogonic lines

(5 × 2) **[10]**

QUESTION 2

Calculate the coordinates of C1, C2 and C3 on ADDENDUM A (attached). Adjustments must be made according to Bowditch's rule. **[20]**

QUESTION 3

A theodolite was set up at station A and readings were taken to spot shots A1, A2, A3 and A4.

Rewrite the information below on ADDENDUM B (attached) and reduce:

- Elevation of point A is 231,99 m.
- Height of instrument at A is 1,43 m.

Readings to spot shots:

POINT	STADIA READINGS	VERTICAL ANGLE	HORIZONTAL READING
A1	2,98……..1,28	83:10:00	14:38:00
A2	3,84……..1,20	92:22:00	96:20:00
A3	3,62……..1,96	99:34:00	206:52:00
A4	2,62..... 1,02	81:46:00	258:32:00

[20]

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QUESTION 4

Calculate the volume of material to be excavated for a channel between points R1 and R2.

The following information is given:

- Chainage of R1: 2 932,82 m
- Chainage of R2: 3 122,24 m
- Formation width: 9 m
- Depth at R1: 2,68 m
- Depth at R2: 1,76 m
- Side slopes: 1:2,5 (1 vertical)
- Ground slope at R1: 1:6 (1 vertical)
- Ground slope at R2: 1:8 (1 vertical)

[14]

QUESTION 5

Explain how to determine volumes from contour lines. **[6]**

QUESTION 6

Combine the following circle-left and circle-right horizontal angle observations which were taken from S1 and find the oriented directions S1–A and S1–C. Use the coordinates of S1 and B.

At S1:

	Circle left	Circle right
A	256:46:28	76:45:48
B	310:56:14	130:56:24
C	88:28:10	268:27:34
A	256:46:12	76:45:28

Coordinates:

S1	-1 467,36	+2 931,21
B	-1 923,26	+2 012,43

[10]

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QUESTION 7

Calculate and tabulate from the information given below the setting-out data for a road curve:

- A peg is required at every full 20 cm chainage.
- The curve is to the left.
- The radius is 210,00 m.
- The angle of intersection Δ is 42:00:00.
- The chainage of the point of intersection (PI) is 4 348,21 m. **[20]**

TOTAL: 100

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ADDENDUM A **EXAMINATION NUMBER:** | | | | | | | | | | | | | |

NAME	JOIN	ΔY	ΔX	NAME	Y	X
A				A	+4 049,94	+6 803,68
209:47:34						
137,78 m						
C1				C1		
126:41:26						
174,36 m						
C2				C2		
99:19:36						
280,46 m						
C3				C3		
214:10:38						
181,70 m						
E				E	+4 296,42	+6 383,32

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ADDENDUM B **EXAMINATION NUMBER:** | | | | | | | | | | | | | |

Station		Distance		HI or middle hair MH	Angles		HI – MH + -	Height component + -	Height difference + -	Elevation of point	Remarks
From	To	Stadia	Hor		Hor	Vert					
A											
	A1										
	A2										
	A3										
	A4										

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FORMULA SHEET

Any other applicable formula may also be used.

$$\alpha = \tan^{-1}\frac{\Delta y}{\Delta x}$$

$$\alpha = \tan^{-1}\frac{\Delta x}{\Delta y} + 90°$$

$$\alpha = \tan^{-1}\frac{\Delta y}{\Delta x} + 180°$$

$$\alpha = \tan^{-1}\frac{\Delta x}{\Delta y} + 270°$$

$$S = \frac{\Delta y}{\sin\alpha}$$

$$S = \frac{\Delta x}{\cos\alpha}$$

$$\Delta y = s.\sin\alpha$$

$$\Delta x = s.\cos\alpha$$

$$C = \frac{\text{Distance}}{\text{Total distance}} X_l$$

$$\Delta h = 50I\sin 2\theta + HI - MH = 100I\sin\theta\cos\theta + HI - MH$$

$$HD = 100I\cos^2\theta$$

$$T = R.\tan\frac{\Delta}{2}$$

$$La = \frac{\pi.\Delta.R}{180}$$

$$\vartheta = \frac{1\,718{,}9.a}{R}$$

$$Cd = T.\tan\frac{\Delta}{4}$$

$$Lc = 2.R.\sin\frac{\Delta}{2}$$

$$W_1 = \frac{g(a+hs)}{g-s}$$

$$W_2 = \frac{g(a+hs)}{g+s}$$

$$A = \frac{W_1.W_2 - a^2}{s}$$

$$a^2 = b^2 + c^2 - 2bc.\cos A$$

$$b^2 = a^2 + c^2 - 2ac.\cos B$$

$$c^2 = a^2 + b^2 - 2ab.\cos C$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

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